
PROFESSIONAL SUMMARY

- Early-career software engineer (MS, Arizona State University, GPA 4.0) who builds full-stack **API services** in **TypeScript (Node.js, Express)** and **Python (FastAPI)** on **PostgreSQL**, and ships **LLM features: tool/function calling and agent workflows** with **Anthropic Claude, LangGraph**, and **AI safety guardrails**, shipping with **Git, CI/CD, Docker**, and **AWS**.

SKILLS

- **LLM & Agentic AI:** LLM APIs (OpenAI, Anthropic, Gemini), LLMs (GPT, Gemini, LLaMA, Qwen), Multi-Agent Architectures, Agentic Workflows & Orchestration Patterns, Tool Calling, MCP-Based Tooling (via Claude Code), LangChain, LangGraph, RAG Pipelines (Chunking, Embeddings, Retrieval Optimization), GraphRAG, Hybrid Search (Vector + Keyword Retrieval), Vector Databases (ChromaDB), Embeddings-Based Retrieval, Knowledge Graphs, Prompt Engineering (CoT, Few-Shot), LLM-as-Judge Evaluation, Faithfulness Verification, Benchmarking (TATQA, MMQA), Transformers (BERT), Ollama
- **Backend & Full Stack:** Node.js, Express, TypeScript, JavaScript, FastAPI, Flask, REST APIs, GraphQL, React, HTML/CSS/JS, Claude Code (AI-assisted coding), Streamlit, NLTK, spaCy, Scrapy
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Hugging Face (Transformers, Accelerate), scikit-learn, XGBoost, CUDA, Deep Learning, NLP, Multimodal AI, Computer Vision, Explainable AI (SHAP, LIME), LoRA/PEFT Fine-Tuning, Threat & Anomaly Detection, Time-Series Modeling, Model Evaluation & Optimization
- **Languages & Databases:** TypeScript, JavaScript, Python, SQL, Java, C++, R, Shell/Bash, PostgreSQL, MySQL, MS SQL Server, MongoDB, DuckDB
- **Testing & Practices:** Unit & Integration Testing, PyTest, Selenium, Cypress, Git & GitHub, GitHub Actions & CI/CD, Code Review, Agile/Scrum, AI Safety Guardrails (Faithfulness Verification, LLM-as-Judge, Prompt/Output Validation)
- **DevOps, Cloud & MLOps:** Docker, Kubernetes, Git/GitHub, GitHub Actions & CI/CD, Azure (Azure ML, Azure DevOps, ADLS), AWS (S3, SageMaker, Redshift), GCP (exposure), Terraform (IaC), MLflow, Apache Airflow, Linux, Power BI, Geospatial/GIS Data (Census, county-level), Grafana, JIRA, Selenium, Cypress
- **Data Engineering & Visualization:** ETL/ELT Pipelines, Data Warehousing, Spark, Hadoop, Parquet, dbt, Snowflake, Tableau, Power BI, Highcharts

EXPERIENCE

- **Arizona State University – Decision Theater Network** Tempe, AZ
Data Analyst – Student Worker May 2025 – Present
 - **Agentic Table-Querying Chatbot:** Architected an LLM-powered chatbot integrating TableGPT with LangGraph-based agentic routing, ChromaDB for vector retrieval, and Streamlit for interactive visualization, deployed via FastAPI for real-time tabular insights.
 - **Exploratory Data Analysis:** Conducted EDA and statistical validation using Python (Pandas, NumPy) to verify model outputs and ensure data quality across diverse datasets.
 - **Visualization Development:** Designed interactive dashboards using Highcharts, Matplotlib, Seaborn, and Tableau; validated LLM-generated visualizations against business requirements and created summary reports in Excel.
 - **Production Inference Optimization:** Optimized TableGPT inference on Intel Gaudi accelerators using BF16/FP16 precision, quantization, and pipeline parallelism, achieving GPU-equivalent latency approximately 90% of the time for production-grade rapid responses.
- **Hewlett Packard Enterprise (Aruba Networking)** Bangalore, India
Cloud Engineer Jan. 2023 – Aug. 2024
 - **Enhanced Aruba's SD-Branch solution:** Engineered production features including traceroute and dynamic DNS for Aruba's SD-Branch platform, improving network reliability by 30%.
 - **Automated Deployment:** Automated scalable infrastructure provisioning and CI/CD workflows using Docker, Kubernetes, YANG models, and Azure DevOps on Linux-based environments.
 - **Workflow Optimization:** Automated REST API and GraphQL scripting workflows, increasing engineering productivity by 50%.
 - **Testing and Quality Assurance:** Reduced critical bugs by 95% through systematic manual and automated testing using JIRA, Selenium, and Cypress.
 - **Agile Methodologies:** Streamlined Agile development processes across cross-functional teams, reducing project delivery time by 20%.
- **Indegene Limited** Bangalore, India
Data Engineer Intern Jun. 2022 – Aug. 2022
 - **Data Migration:** Ensured 100% data integrity during migrations from SFTP, Azure, and SFMC platforms using SQL Server stored procedures, improving data reliability and system performance.
 - **ETL and Big Data Processing:** Engineered data pipelines orchestrated with Apache Airflow using Python, Spark, and Hadoop, reducing processing time by 30% and enabling efficient stakeholder reporting.
 - **Data Warehousing:** Optimized AWS S3 and Redshift workflows with PySpark, improving transfer efficiency by 25%; monitored pipeline health using Grafana dashboards.

EDUCATION

- **Arizona State University** Arizona, AZ
Master of Science in Data Science, Analytics and Engineering; GPA: 4.0 *Aug. 2024 – May. 2026*
- **PES University** Bangalore, India
Bachelor of Technology in Computer Science; GPA: 3.68 (9.06/10.0) *Aug. 2019 – May. 2023*

PROJECTS

- **RateMyRep HackASU 2026 AI & Agents Engineer:** Built an agent-driven pipeline using **Claude Code**, Gemini 2.5 Flash, and TinyFish API to extract, classify, and reconcile legislator data from bills, voter guides, and public statements. Used Claude for PDF parsing and statement consolidation from Arizona Clean Elections voter guides, and designed prompting workflows for Gemini to perform batched topic classification across 12 policy categories. Integrated TinyFish as a web-scraping agent to gather public statements for 30 representatives, with PostgreSQL-backed storage and backend services deployed through InsForge.
- **Multi-Agent Framework for Context-Aware Table Summarization:** A multi-agent LLM pipeline using LangChain and LangGraph to orchestrate specialized models across modalities TableGPT for tabular QA, Gemini for text QA, BLIP2 for image captioning, and Gemma for summary generation with LoRA fine-tuning via Hugging Face Accelerate. Applied Chain-of-Thought reasoning, instruction prompting, and faithfulness verification to improve output quality and reduce hallucinated summaries. Evaluated on TATQA and MMQA benchmarks using an LLM-as-Judge pipeline measuring answer relevance, grounding, and summary faithfulness.
- **Hospital Price Transparency QA System:** Developed an AI-powered question-answering system over healthcare pricing data, combining **hybrid search** (semantic vector retrieval via ChromaDB with keyword and SQL filtering) and GraphRAG for procedure similarity search across 120 Arizona hospitals and 97 MRF files. Integrated TableGPT for tabular question answering and DuckDB for SQL-based price analysis. Applied prompt engineering with Qwen LLM to enable natural language queries about procedure pricing, plan comparisons, and cross-hospital cost analysis. Automated data collection via Scrapy and structured 4GB of pricing data for AI-driven insights.
- **PennyPath: AI Financial Planning Platform:** Developed an AI-driven financial planning platform using GraphRAG, a knowledge graph built from verified financial documents, using Gemini based personalization to deliver user-specific recommendations. Designed a hybrid pipeline combining persona-aware retrieval, deterministic rule-based prioritization, and interactive graph-backed explanations so users could see why plans were ordered a certain way. Improved responsiveness by structuring What-If scenarios to re-run only the Financial Order of Operations, cutting update latency to ~50ms and supporting real-time interaction through Flask APIs.
- **E-Commerce Platform (MERN Full-Stack):** Built a full-stack e-commerce application with **Node.js, Express, React, and MongoDB**, implementing user authentication, product catalog, shopping cart, and checkout flows over **REST APIs**. Designed the backend service layer and API endpoints for secure, stateful interactions between the front-end and data store, and structured the codebase for maintainability and iterative feature delivery.
- **Threat Detection with Explainable AI (AI Safety / Anomaly Checks):** Built an explainable security-ML pipeline for threat and anomaly detection using **XGBoost** and scikit-learn, improving prediction accuracy by 20% over a baseline. Integrated **SHAP** and **LIME** to explain decisions and surface highest-risk signals, an approach directly relevant to guardrails such as data-exposure and prompt-injection detection.
- **CMI BFRB Detection (Wearable / IoT Multimodal Sensor Time-Series):** Built a multimodal time-series classification pipeline over wrist-worn IoT sensor streams (IMU, thermopile, ToF) to detect Body-Focused Repetitive Behaviors. Implemented sequence windowing, normalization, and missing-value handling, then trained a Conv1D model reaching 96.02% Binary F1.

PUBLICATION

Kulkarni, R. P., & Puttane, S. M. (2024). Framework for Home Layout Design by Semantic Parsing of Text. *Advances in Computational Intelligence and Its Applications*, 195.